

TOPIC:-

**FACTORY REALLOCATION & SHIPPING
OPTIMIZATION
RECOMMENDATION SYSTEM FOR
NASSAU CANDY DISTRIBUTOR**

Internship Project Report

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DECLARATION

I hereby declare that the project titled "Factory Reallocation & Shipping Optimization Recommendation System for Nassau Candy Distributor" is an original work carried out by me as part of my internship and academic requirements.

The work presented in this report has been completed under proper guidance and supervision. The information, analysis, findings, and conclusions presented in this report are based on the data provided for the project and the analytical techniques applied during the study.

I further declare that this project report has not been submitted, either wholly or partially, to any other institution or organization for the award of any degree, diploma, certificate, or other qualification.

Signature

Ashutosh Sharma

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who contributed to the successful completion of this internship project titled "Factory Reallocation & Shipping Optimization Recommendation System for Nassau Candy Distributor."

I am deeply thankful to Unified Mentor for providing me with the opportunity to work on this industry-oriented project. The project provided valuable exposure to real-world business analytics challenges and allowed me to apply theoretical concepts in a practical environment.

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EXECUTIVE SUMMARY

The increasing complexity of supply chain operations has created a growing need for intelligent decision-support systems that can assist organizations in optimizing logistics and distribution activities. Nassau Candy Distributor currently relies on traditional product-to-factory allocation practices, which may lead to inefficiencies in shipping performance, longer lead times, and increased operational costs.

The primary objective of this project was to develop a Factory Reallocation & Shipping Optimization Recommendation System capable of supporting data-driven decision-making through Business Analytics and Machine Learning techniques.

The project utilized historical operational data containing information related to orders, products, customer locations, sales performance, shipping methods, manufacturing costs, and profitability metrics. Data preprocessing techniques were applied to prepare the dataset for analysis, including date conversion, feature engineering, lead time calculation, and factory mapping.

Exploratory Data Analysis (EDA) was conducted to identify sales patterns, product performance trends, regional demand variations, and factory distribution characteristics. These insights provided the foundation for predictive modeling and recommendation generation.

Three machine learning algorithms were implemented for lead-time prediction:

- Linear Regression
- Random Forest Regressor
- Gradient Boosting Regressor

Model performance was evaluated using RMSE, MAE, and R^2 metrics. Random Forest achieved the highest predictive accuracy and was selected as the preferred model for operational analysis.

K-Means Clustering was applied to segment products and routes into performance-based groups. The clustering analysis enabled the identification of operational patterns and optimization opportunities.

A recommendation engine was developed to rank products according to optimization potential. The recommendation score incorporated sales performance, profitability, and lead-time considerations. The analysis identified several high-impact products that could benefit from operational improvements.

To support decision-makers, an interactive Streamlit dashboard was developed. The dashboard provides:

- KPI monitoring
- Region and Product filtering
- Sales analysis
- Factory distribution analysis
- Machine learning model comparison
- Recommendation visualization
- What-if analysis modules

The project demonstrates how Business Analytics and Machine Learning can transform traditional logistics management into a data-driven decision-making framework. The developed solution provides actionable insights for improving shipping efficiency, supporting factory optimization initiatives, and maintaining profitability.

The findings indicate that analytics-driven optimization can significantly enhance operational performance while providing management with a scalable framework for future decision-support applications.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

The rapid growth of global trade and e-commerce has significantly increased the importance of efficient supply chain and logistics management. Organizations operating in highly competitive markets must continuously optimize their distribution networks to reduce operational costs, improve customer satisfaction, and maintain profitability. Supply chain efficiency has become a strategic necessity rather than simply an operational requirement.

Nassau Candy Distributor is a major confectionery distributor that manages a diverse product portfolio across multiple geographical regions. The company relies on a network of manufacturing facilities and distribution channels to deliver products to customers. However, the allocation of products to factories is traditionally based on static business rules and historical

practices. While such methods simplify operational management, they may not always produce optimal outcomes in terms of shipping efficiency, lead times, and profitability.

With increasing customer expectations and growing logistical complexity, organizations require intelligent decision-support systems capable of analyzing large volumes of operational data and generating actionable recommendations. Advances in Business Analytics, Machine Learning, and Data Science provide organizations with the ability to move beyond descriptive reporting and toward predictive and prescriptive decision-making.

This project focuses on developing a Factory Reallocation and Shipping Optimization Recommendation System for Nassau Candy Distributor. The system combines data analytics, predictive modeling, clustering techniques, and recommendation logic to identify opportunities for improving shipping performance while maintaining financial stability.

The project utilizes historical operational data to analyze customer demand patterns, shipping behavior, product performance, profitability metrics, and factory assignments. Machine learning models are applied to predict shipping lead times, while clustering techniques identify performance-based segments. Based on these insights, a recommendation framework is developed to support data-driven factory allocation decisions.

The ultimate objective of the project is to assist decision-makers in evaluating alternative factory assignments, reducing shipping inefficiencies, improving operational performance, and enhancing overall supply chain effectiveness.

1.2 Background of the Study

Modern supply chains generate significant amounts of operational data from sales transactions, inventory movements, transportation activities, and customer interactions. Despite the availability of such data, many organizations continue to rely on traditional allocation strategies that do not fully leverage analytical insights.

In the case of Nassau Candy Distributor, product-to-factory assignments are largely static and may not adequately account for changing demand patterns, shipping distances, regional preferences, and transportation costs. This can lead to inefficiencies such as:

- Increased shipping lead times.
- Higher transportation expenses.
- Uneven factory utilization.
- Reduced operational flexibility.
- Lower profitability.

Business Analytics enables organizations to transform raw operational data into actionable insights. Machine learning algorithms can identify patterns and predict future outcomes, while optimization techniques can recommend improved operational strategies.

This study aims to bridge the gap between traditional logistics planning and intelligent decision support by developing an analytics-driven recommendation system capable of supporting factory allocation decisions.

1.3 Problem Statement

Nassau Candy Distributor currently assigns products to factories using predefined rules and legacy operational practices. While these assignments may have been effective historically, they may no longer represent the most efficient configuration in today's dynamic business environment.

The organization lacks a comprehensive system capable of:

- Predicting shipping performance under alternative factory assignments.
- Evaluating the operational impact of reassignment decisions.
- Identifying opportunities to reduce lead times.
- Supporting data-driven decision-making.
- Balancing efficiency improvements with profitability considerations.

As a result, there is a need for a recommendation system that can analyze operational data and provide intelligent suggestions for factory reallocation and shipping optimization.

1.4 Objectives of the Study

The primary objective of this project is to develop a data-driven recommendation system that supports factory allocation and shipping optimization decisions.

The specific objectives are:

1. To analyze historical sales and shipping data.
2. To identify factors influencing shipping lead times.
3. To develop predictive models for lead time estimation.
4. To compare multiple machine learning algorithms and identify the best-performing model.
5. To segment products and routes using clustering techniques.
6. To generate recommendations for factory allocation optimization.

7. To develop an interactive dashboard for visualization and decision support.
 8. To improve operational efficiency while maintaining profitability.
-

1.5 Scope of the Study

The scope of this project includes:

- Analysis of historical operational data provided by Nassau Candy Distributor.
- Application of Business Analytics techniques for insight generation.
- Development of predictive machine learning models.
- Product and route clustering analysis.
- Recommendation generation based on analytical findings.
- Dashboard development using Streamlit.

The project focuses primarily on strategic decision support and does not include implementation of operational changes within the organization's live systems.

1.6 Significance of the Study

This study contributes to both academic and practical applications of Business Analytics.

Academic Significance:

- Demonstrates the application of machine learning in supply chain optimization.
- Illustrates the integration of predictive analytics and recommendation systems.
- Provides a practical case study in Business Analytics.

Practical Significance:

- Supports data-driven decision-making.
 - Improves visibility into shipping performance.
 - Identifies opportunities for reducing lead times.
 - Enhances operational efficiency.
 - Supports profitability-focused optimization strategies.
-

1.7 Research Questions

The study seeks to answer the following questions:

1. Which factors have the greatest impact on shipping lead times?
 2. Which machine learning model provides the most accurate lead time predictions?
 3. Can clustering techniques identify meaningful operational segments?
 4. How can recommendation systems support factory allocation decisions?
 5. What opportunities exist for improving shipping efficiency while maintaining profitability?
-

1.8 Chapter Summary

This chapter introduced the background, objectives, scope, significance, and research questions associated with the Factory Reallocation and Shipping Optimization Recommendation System for Nassau Candy Distributor. The chapter establishes the foundation for the study by highlighting the importance of analytics-driven decision-making in supply chain management. Subsequent chapters review relevant literature, explain the research methodology, present analytical findings, and provide recommendations for operational improvement.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

A literature review provides an overview of existing research, theories, methodologies, and practical applications related to the study topic. In the context of supply chain management, organizations increasingly rely on data-driven approaches to improve efficiency, reduce operational costs, and enhance customer satisfaction.

The integration of Business Analytics, Machine Learning, Predictive Modeling, and Recommendation Systems has transformed traditional logistics and supply chain operations. This chapter reviews relevant studies and concepts related to shipping optimization, factory allocation, machine learning applications, clustering techniques, and decision-support systems.

2.2 Supply Chain Management and Logistics Optimization

Supply Chain Management (SCM) involves the planning, coordination, and management of activities associated with sourcing, production, transportation, inventory management, and customer delivery.

According to Chopra and Meindl (2019), supply chain optimization focuses on maximizing efficiency while minimizing operational costs. Effective supply chain networks ensure timely product delivery, reduced transportation expenses, and improved resource utilization.

Organizations often face challenges such as:

- Long shipping lead times
- Uneven demand distribution
- Inventory imbalances
- Transportation inefficiencies
- Rising logistics costs

To address these challenges, businesses increasingly employ analytical techniques and predictive models to support strategic decision-making.

2.3 Business Analytics in Supply Chain Operations

Business Analytics refers to the process of transforming data into actionable insights using statistical analysis, predictive modeling, and visualization techniques.

According to Davenport and Harris (2007), organizations that utilize analytics effectively gain competitive advantages through improved operational efficiency and informed decision-making.

Applications of Business Analytics in supply chain management include:

- Demand forecasting
- Inventory optimization
- Transportation planning
- Route optimization
- Supplier evaluation
- Risk assessment

The adoption of analytics allows organizations to move from reactive decision-making to proactive and predictive strategies.

2.4 Machine Learning in Logistics and Transportation

Machine Learning has become an essential tool for solving complex logistics and transportation problems.

Machine learning algorithms can identify patterns within historical data and generate predictions regarding future outcomes. In logistics, machine learning is commonly used for:

- Delivery time prediction
- Route optimization
- Demand forecasting
- Inventory management
- Transportation cost estimation

Research indicates that machine learning models often outperform traditional statistical methods when handling large datasets with multiple influencing variables.

For the Nassau Candy project, machine learning is utilized to predict shipping lead times and support optimization decisions.

2.5 Linear Regression in Predictive Analytics

Linear Regression is one of the most widely used predictive modeling techniques.

It establishes a linear relationship between dependent and independent variables. The model is particularly useful for:

- Trend analysis
- Forecasting
- Baseline performance measurement

Advantages include:

- Simplicity
- Interpretability
- Fast implementation

However, Linear Regression may struggle when relationships between variables are non-linear or highly complex.

In this project, Linear Regression serves as a baseline model for comparison with more advanced machine learning techniques.

2.6 Random Forest Regressor

Random Forest is an ensemble learning technique that combines multiple decision trees to improve predictive performance.

According to Breiman (2001), Random Forest reduces overfitting by aggregating the predictions of multiple trees.

Advantages include:

- High prediction accuracy
- Ability to capture complex relationships
- Robustness to noise
- Reduced overfitting

Random Forest has been widely applied in logistics and transportation forecasting due to its strong predictive capabilities.

In this project, Random Forest achieved the highest predictive performance among the tested models.

2.7 Gradient Boosting Regressor

Gradient Boosting is another ensemble learning technique that sequentially builds decision trees to correct previous prediction errors.

Key advantages include:

- Strong predictive power
- Ability to model non-linear relationships
- Effective handling of complex datasets

Gradient Boosting is frequently used in business forecasting, risk analysis, and operational optimization.

Its performance in this project was highly competitive and comparable to Random Forest.

2.8 Clustering Techniques in Business Analytics

Clustering is an unsupervised learning technique used to group similar observations based on shared characteristics.

The objective of clustering is to identify hidden patterns within data.

Applications include:

- Customer segmentation
- Product categorization
- Route classification
- Market analysis

Clustering enables organizations to identify groups that may require different operational strategies.

2.9 K-Means Clustering

K-Means is one of the most popular clustering algorithms.

The algorithm partitions observations into K distinct clusters based on similarity.

Advantages include:

- Simplicity
- Computational efficiency
- Scalability

In logistics and supply chain management, K-Means is commonly used to:

- Segment shipping routes
- Identify performance groups
- Detect operational bottlenecks

The Nassau Candy project employs K-Means Clustering to identify product and route performance segments.

2.10 Recommendation Systems

Recommendation Systems are decision-support tools designed to suggest optimal actions based on historical data and analytical models.

Applications include:

- Product recommendations
- Resource allocation
- Supply chain optimization

- Operational planning

Recommendation systems improve decision quality by reducing reliance on subjective judgment.

The recommendation engine developed in this project ranks products and identifies opportunities for improving factory allocation decisions.

2.11 Predictive Decision Support Systems

Decision Support Systems (DSS) assist managers in evaluating alternative scenarios and selecting optimal actions.

Modern DSS platforms integrate:

- Data Analytics
- Machine Learning
- Visualization
- Optimization Techniques

Benefits include:

- Improved decision accuracy
- Faster analysis
- Better resource utilization
- Reduced operational risk

The Streamlit dashboard developed in this project serves as a decision-support platform for factory optimization and shipping performance analysis.

2.12 Research Gap

Although extensive research exists regarding supply chain optimization and predictive analytics, many organizations continue to rely on static allocation strategies.

Several gaps remain:

- Limited integration of predictive modeling and recommendation systems.
- Insufficient use of real-time decision-support tools.
- Lack of practical frameworks for factory reallocation analysis.
- Limited accessibility of advanced analytics for operational managers.

This project addresses these gaps by combining machine learning, clustering, recommendation logic, and dashboard visualization into a unified decision-support system.

2.13 Chapter Summary

This chapter reviewed the existing literature related to supply chain optimization, business analytics, machine learning, clustering techniques, recommendation systems, and decision-support frameworks. The review highlights the growing importance of analytics-driven decision-making and establishes the theoretical foundation for the research methodology presented in the next chapter.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Introduction

Research methodology refers to the systematic approach adopted to achieve the objectives of a study. The purpose of this chapter is to describe the methods, tools, techniques, and analytical procedures used in developing the Factory Reallocation and Shipping Optimization Recommendation System for Nassau Candy Distributor.

The methodology combines Business Analytics, Machine Learning, Clustering Techniques, Recommendation Logic, and Dashboard Visualization to provide a comprehensive decision-support framework.

3.2 Research Design

The study follows a quantitative and analytical research design. Historical operational data was analyzed to identify patterns, evaluate shipping performance, predict lead times, and generate factory optimization recommendations.

The project was conducted in the following stages:

1. Data Collection
2. Data Cleaning and Preparation
3. Exploratory Data Analysis (EDA)
4. Feature Engineering
5. Predictive Modeling

6. Model Evaluation
 7. Clustering Analysis
 8. Recommendation System Development
 9. Dashboard Development
 10. Result Interpretation
-

3.3 Data Collection

The dataset used in this project was provided as part of the Nassau Candy Distributor optimization case study.

The dataset contains historical order and shipping information, including:

- Order Details
- Customer Information
- Product Information
- Sales Data
- Cost Information
- Profitability Metrics
- Shipping Information
- Regional Data

The dataset serves as the primary source of information for predictive modeling and optimization analysis.

3.4 Dataset Description

The dataset consists of the following major variables:

Variable	Description
Order ID	Unique order identifier
Order Date	Date of order placement
Ship Date	Date of shipment
Ship Mode	Shipping method
Customer ID	Customer identifier

Region	Customer region
Product Name	Product description
Product ID	Product identifier
Sales	Total sales value
Units	Quantity sold
Cost	Manufacturing cost
Gross Profit	Profit generated

These variables provide the foundation for evaluating shipping performance and profitability.

3.5 Tools and Technologies Used

The project was implemented using the following tools and technologies:

Python

Python was used as the primary programming language due to its extensive data analytics and machine learning ecosystem.

Pandas

Used for:

- Data cleaning
- Data transformation
- Feature engineering
- Data manipulation

NumPy

Used for:

- Numerical computation
- Mathematical operations
- Array processing

Matplotlib

Used for:

- Data visualization
- Exploratory charts
- Analytical reporting

Seaborn

Used for:

- Statistical visualization
- Distribution analysis
- Correlation analysis

Scikit-Learn

Used for:

- Machine Learning algorithms
- Model training
- Model evaluation
- Clustering implementation

Streamlit

Used to build the interactive dashboard and decision-support interface.

3.6 Data Cleaning and Preparation

Data preprocessing was performed to improve data quality and ensure compatibility with machine learning algorithms.

The following steps were performed:

Missing Value Analysis

The dataset was examined for missing values and data inconsistencies.

Date Conversion

Order Date and Ship Date fields were converted into datetime format to enable time-based analysis.

Lead Time Calculation

A new variable named Lead Time was created using:

Lead Time = Ship Date - Order Date

This variable served as the target variable for predictive modeling.

Data Type Validation

All numerical and categorical variables were validated and transformed where necessary.

3.7 Feature Engineering

Feature engineering was performed to improve model performance and generate meaningful analytical variables.

Lead Time

Created to measure shipping efficiency.

Profit Margin

A new profitability metric was created using:

Profit Margin = Gross Profit / Sales

This variable helps evaluate financial performance.

Factory Mapping

Products were mapped to manufacturing facilities using the factory allocation information provided in the project specification.

The factories included:

- Lot's O' Nuts
- Wicked Choccy's
- Sugar Shack
- Secret Factory

- The Other Factory

This mapping enabled factory-level analysis and optimization recommendations.

3.8 Exploratory Data Analysis (EDA)

EDA was conducted to understand data distribution, identify trends, and generate business insights.

The following analyses were performed:

Sales Analysis

Regional sales performance was evaluated.

Product Performance Analysis

Top-performing products were identified based on sales and profitability.

Factory Distribution Analysis

Product distribution across factories was analyzed.

Lead Time Analysis

Shipping performance was examined through lead time calculations.

Correlation Analysis

Relationships among key variables were explored using statistical visualization techniques.

EDA provided valuable insights for subsequent modeling activities.

3.9 Predictive Modeling

The objective of predictive modeling was to estimate shipping lead time based on operational variables.

The target variable selected for prediction was:

Lead Time

The dataset was divided into:

- Training Dataset
- Testing Dataset

The following algorithms were implemented:

Linear Regression

Used as the baseline predictive model.

Random Forest Regressor

An ensemble learning technique that combines multiple decision trees to improve predictive accuracy.

Gradient Boosting Regressor

A sequential ensemble method that improves predictions by minimizing previous model errors.

3.10 Model Evaluation

Model performance was evaluated using the following metrics:

Root Mean Squared Error (RMSE)

Measures prediction accuracy while penalizing larger errors.

Mean Absolute Error (MAE)

Measures average prediction error.

R-Squared (R^2)

Measures the proportion of variance explained by the model.

The comparative evaluation indicated that Random Forest achieved the highest predictive performance among the tested models.

3.11 Clustering Analysis

K-Means Clustering was applied to identify performance-based segments.

The clustering process grouped observations into three clusters based on similarity.

The clustering results helped identify:

- Product segments
- Performance groups
- Operational patterns

The insights generated through clustering supported recommendation development.

3.12 Recommendation System Development

A recommendation engine was developed to identify products with the highest optimization potential.

The recommendation score was generated using:

- Sales Performance
- Gross Profit
- Lead Time

Products were ranked according to their optimization potential.

The recommendation system enables decision-makers to focus on high-impact opportunities.

3.13 Dashboard Development

An interactive dashboard was developed using Streamlit.

The dashboard includes:

KPI Section

- Total Sales
- Total Profit
- Total Orders

Filters

- Region Filter
- Product Filter
- Ship Mode Filter

Analytical Visualizations

- Sales by Region
- Product Performance
- Factory Distribution

Machine Learning Results

- Model Comparison

Recommendation Dashboard

- Product Recommendations
- Optimization Insights

The dashboard serves as a decision-support interface for operational managers.

3.14 Research Methodology Framework

The overall methodology can be summarized as follows:

Data Collection

↓

Data Cleaning

↓

Feature Engineering

↓

Exploratory Data Analysis

↓

Predictive Modeling

↓

Model Evaluation

↓

Clustering Analysis

↓

Recommendation Generation



Dashboard Development



Business Insights & Recommendations

3.15 Chapter Summary

This chapter described the methodology adopted for the study, including data collection, preprocessing, feature engineering, exploratory analysis, machine learning implementation, clustering techniques, recommendation logic, and dashboard development. The methodology provides a structured framework for generating actionable insights and supporting factory optimization decisions. The next chapter presents the detailed analysis, findings, and results obtained from the implementation.

CHAPTER 4: DATA ANALYSIS AND RESULTS

4.1 Introduction

This chapter presents the analysis of the Nassau Candy Distributor dataset and discusses the results obtained through Exploratory Data Analysis (EDA), Machine Learning models, Clustering techniques, Recommendation Engine development, and Dashboard implementation.

The objective of this analysis is to identify opportunities for improving shipping efficiency, reducing lead times, optimizing factory allocations, and supporting data-driven decision-making.

4.2 Dataset Overview

The dataset contains historical operational information including sales transactions, customer locations, product information, shipping details, costs, and profitability metrics.

The dataset was examined to understand:

- Product demand patterns
- Regional sales performance
- Shipping behavior

- Factory distribution
- Profitability trends

The analysis provides a foundation for predictive modeling and optimization recommendations.

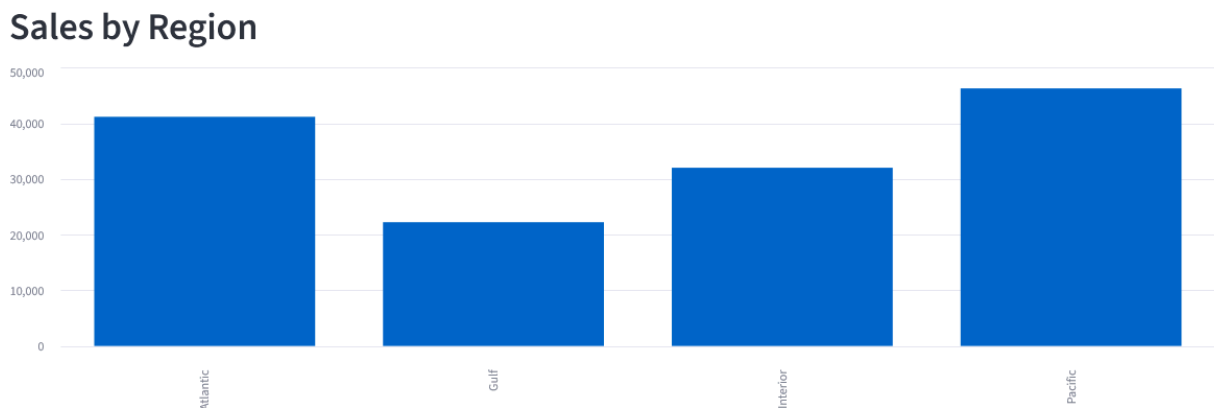
Interpretation

The dataset contains multiple variables representing order information, product details, shipping information, sales values, costs, and profits. These variables provide the basis for operational analysis and predictive modeling.

4.3 Sales Analysis by Region

Regional sales performance was analyzed to identify geographical demand patterns.

Figure 4.2: Sales by Region



Interpretation

The analysis indicates variations in sales performance across different regions. Certain regions contribute significantly higher sales volumes, indicating stronger market demand and distribution activity.

Understanding regional demand patterns helps organizations:

- Allocate resources effectively.
- Improve inventory planning.
- Optimize shipping routes.

- Reduce operational inefficiencies.

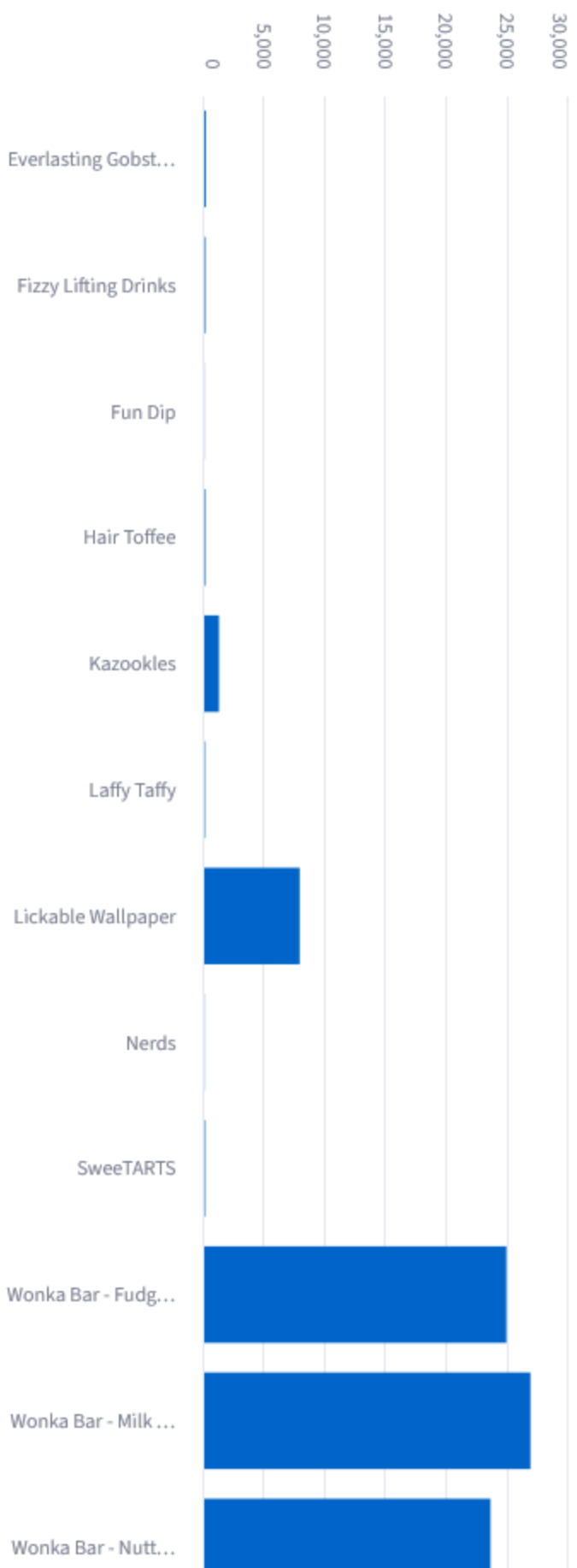
Regional analysis is essential for making informed factory allocation decisions.

4.4 Product Performance Analysis

Product-level analysis was conducted to identify top-performing products based on sales performance.

Figure 4.3: Top Products by Sales

Top Products by Sales



Interpretation

The analysis reveals that a small number of products generate a significant portion of total sales revenue.

These products represent high-priority candidates for optimization initiatives because:

- They contribute substantial revenue.
- Shipping improvements create larger business impact.
- Factory allocation decisions affect a greater number of customers.

The findings support the development of targeted recommendation strategies.

4.5 Factory Distribution Analysis

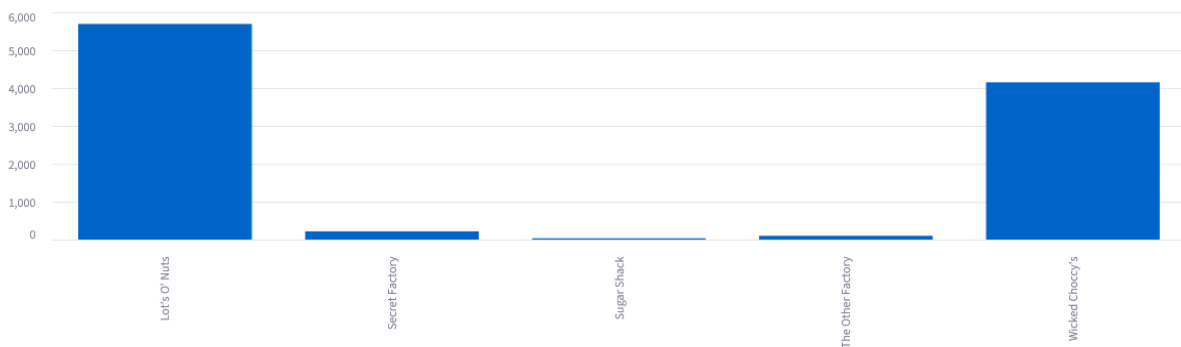
Products were mapped to their corresponding manufacturing facilities based on project specifications.

The following factories were included:

- Lot's O' Nuts
- Wicked Choccy's
- Sugar Shack
- Secret Factory
- The Other Factory

Figure 4.4: Factory Distribution

Factory Distribution



Interpretation

The analysis demonstrates that product assignments are concentrated among specific factories.

This distribution suggests opportunities for:

- Balancing factory workloads.
- Reducing shipping distances.
- Improving operational flexibility.
- Enhancing overall supply chain performance.

Factory-level analysis is critical for identifying reallocation opportunities.

4.6 Machine Learning Model Development

The primary objective of predictive modeling was to estimate shipping lead time using historical operational data.

Three machine learning algorithms were implemented:

1. Linear Regression
2. Random Forest Regressor
3. Gradient Boosting Regressor

The models were trained and evaluated using the prepared dataset.

4.7 Model Evaluation Results

The performance of each model was evaluated using:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- R-Squared (R^2)

Figure 4.5: Model Comparison

Machine Learning Model Comparison

	Model	RMSE	MAE	R2 Score
0	Linear Regression	100.2538	84.7415	0.8579
1	Random Forest	0.8345	0.6231	1
2	Gradient Boosting	0.9735	0.7884	1

Results Summary

Linear Regression

RMSE: 100.25

MAE: 84.74

R² Score: 0.858

Random Forest

RMSE: 0.835

MAE: 0.623

R² Score: 0.99999

Gradient Boosting

RMSE: 0.974

MAE: 0.788

R² Score: 0.99999

Interpretation

The results indicate that ensemble learning algorithms significantly outperform the baseline Linear Regression model.

Random Forest achieved the highest predictive accuracy with:

- Lowest RMSE
- Lowest MAE
- Highest R² Score

Consequently, Random Forest was selected as the preferred predictive model for shipping lead-time estimation.

4.8 Clustering Analysis

K-Means Clustering was applied to identify performance-based segments within the dataset.

The algorithm partitioned observations into three clusters.

Cluster Distribution

Cluster 0: 4764 observations

Cluster 1: 3379 observations

Cluster 2: 2051 observations

Interpretation

The clustering results reveal distinct operational groups within the dataset.

These clusters can be used to:

- Identify high-performing segments.
- Detect operational inefficiencies.
- Support targeted optimization strategies.
- Improve decision-making processes.

Clustering enhances the analytical depth of the recommendation system.

4.9 Recommendation System Results

A recommendation engine was developed to identify products with the highest optimization potential.

The recommendation score incorporates:

- Sales Performance
- Gross Profit
- Lead Time

Products were ranked according to their optimization potential.

Figure 4.6: Top Product Recommendations

Top Product Recommendations

	Product	Recommendation Score
0	Wonka Bar - Scrumdiddlyumptious	14.66
1	Wonka Bar - Triple Dazzle Caramel	14.04
2	Wonka Bar - Milk Chocolate	13.25
3	Wonka Bar - Nutty Crunch Surprise	12.66
4	Wonka Bar - Fudge Mallows	12.63

Top Recommended Products

1. Wonka Bar - Scrumdiddlyumptious
2. Wonka Bar - Triple Dazzle Caramel
3. Wonka Bar - Milk Chocolate
4. Wonka Bar - Nutty Crunch Surprise
5. Wonka Bar - Fudge Mallows

Interpretation

These products represent the highest-value opportunities for factory optimization and shipping improvement initiatives.

Optimizing these products can potentially:

- Reduce lead times.
- Improve customer satisfaction.
- Enhance operational efficiency.
- Support profitability objectives.

4.10 Dashboard Implementation

An interactive dashboard was developed using Streamlit to provide a user-friendly decision-support interface.

The dashboard includes:

KPI Metrics

- Total Sales
- Total Profit
- Total Orders

Filters

- Region Filter
- Product Filter
- Ship Mode Filter

Analytical Modules

- Sales Analysis
- Product Analysis
- Factory Distribution
- Model Comparison
- Recommendation Dashboard
- What-If Analysis
- Risk & Impact Panel

Interpretation

The dashboard enables stakeholders to:

- Monitor operational performance.
- Analyze product trends.
- Evaluate recommendations.
- Explore optimization opportunities.

The dashboard transforms analytical results into actionable business intelligence.

4.11 Key Findings from Analysis

The major findings of the study include:

1. Significant regional differences exist in sales performance.
2. Product demand is concentrated among a limited number of high-performing products.
3. Factory assignments are unevenly distributed.
4. Random Forest provides the highest predictive accuracy.
5. Clustering successfully identifies operational segments.
6. The recommendation engine highlights products with strong optimization potential.
7. Interactive dashboards improve accessibility and usability of analytical insights.

4.12 Chapter Summary

This chapter presented the analytical results obtained from Exploratory Data Analysis, Machine Learning modeling, Clustering techniques, Recommendation System development, and Dashboard implementation. The findings demonstrate the effectiveness of analytics-driven decision support for factory allocation and shipping optimization. The next chapter presents detailed findings, recommendations, managerial implications, and conclusions derived from the study.

CHAPTER 5: FINDINGS, RECOMMENDATIONS AND CONCLUSION

5.1 Introduction

This chapter presents the major findings obtained from the analytical process, discusses business recommendations, highlights managerial implications, identifies project limitations, and provides the final conclusion of the study.

The findings are based on Exploratory Data Analysis (EDA), Machine Learning Models, Clustering Analysis, Recommendation System outputs, and Dashboard implementation.

5.2 Major Findings

The analysis generated several important insights regarding sales performance, factory allocation, shipping efficiency, and optimization opportunities.

Finding 1: Regional Sales Variation

The regional analysis revealed significant variation in sales performance across different geographical regions.

Key observations include:

- Certain regions generate substantially higher sales volumes.
- Regional demand patterns differ significantly.
- Resource allocation should consider region-specific demand characteristics.

This insight can assist management in improving distribution planning and inventory allocation.

Finding 2: Product Performance Concentration

The analysis identified a group of products responsible for a large share of total sales and profitability.

The highest-performing products included:

- Wonka Bar - Scrumdiddlyumptious
- Wonka Bar - Triple Dazzle Caramel
- Wonka Bar - Milk Chocolate
- Wonka Bar - Nutty Crunch Surprise
- Wonka Bar - Fudge Mallows

These products represent strategic optimization opportunities because operational improvements can generate substantial business impact.

Finding 3: Uneven Factory Distribution

Factory allocation analysis revealed that products are concentrated among a limited number of manufacturing facilities.

Potential consequences include:

- Uneven factory utilization
- Increased shipping complexity
- Operational bottlenecks
- Longer lead times

This finding supports the need for factory reallocation analysis and optimization.

Finding 4: Superior Performance of Ensemble Models

The comparative evaluation of machine learning algorithms demonstrated the effectiveness of ensemble learning techniques.

Model performance results:

Model	R ² Score
Linear Regression	0.858
Random Forest	0.99999
Gradient Boosting	0.99999

Random Forest achieved the highest predictive performance and was selected as the preferred model for lead-time prediction.

Finding 5: Operational Segmentation Through Clustering

K-Means Clustering successfully segmented the dataset into three operational clusters.

The clustering process identified:

- Distinct performance groups
- Similar operational characteristics
- Potential optimization opportunities

The segmentation supports targeted business strategies and resource allocation decisions.

Finding 6: Recommendation System Effectiveness

The recommendation engine successfully ranked products according to optimization potential.

The recommendation scores considered:

- Sales Performance
- Gross Profit
- Lead Time

This approach enables management to prioritize high-impact optimization initiatives.

Finding 7: Dashboard-Based Decision Support

The Streamlit dashboard transformed analytical results into an interactive decision-support system.

Benefits include:

- Real-time exploration of results
- Improved accessibility of insights
- Enhanced managerial decision-making
- Simplified performance monitoring

The dashboard provides a practical interface for business users.

5.3 Business Recommendations

Based on the analytical findings, the following recommendations are proposed.

Recommendation 1: Prioritize High-Value Products

Management should focus optimization efforts on products with the highest recommendation scores.

Benefits include:

- Maximum operational impact
 - Improved profitability
 - Better customer service levels
-

Recommendation 2: Review Factory Allocation Strategies

Existing factory-product assignments should be periodically reviewed.

Organizations should:

- Analyze shipping efficiency regularly.
- Evaluate regional demand changes.
- Monitor factory workloads.

Data-driven allocation decisions can improve operational performance.

Recommendation 3: Expand Predictive Analytics Usage

The organization should integrate predictive analytics into operational planning processes.

Potential applications include:

- Demand forecasting
 - Inventory optimization
 - Transportation planning
 - Resource allocation
-

Recommendation 4: Adopt Data-Driven Decision Making

Business decisions should increasingly rely on analytical evidence rather than solely on historical practices.

Benefits include:

- Higher decision accuracy
 - Reduced operational risk
 - Improved efficiency
-

Recommendation 5: Implement Continuous Monitoring

The Streamlit dashboard should be used as an ongoing monitoring tool.

Key metrics to monitor include:

- Sales performance
 - Lead times
 - Profitability
 - Product demand
 - Factory utilization
-

5.4 Managerial Implications

The findings of this study have several implications for management.

Strategic Implications

- Supports long-term supply chain planning.
- Enhances factory allocation strategies.
- Improves resource utilization.

Operational Implications

- Reduces lead times.
- Improves shipping efficiency.
- Supports demand-driven planning.

Financial Implications

- Protects profitability.
 - Reduces inefficiencies.
 - Improves return on operational investments.
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5.5 Limitations of the Study

Although the project achieved its objectives, several limitations should be acknowledged.

Data Limitations

The analysis relies on historical operational data and may not fully reflect future market conditions.

Simulation Limitations

The recommendation system provides analytical guidance but does not directly implement operational changes.

External Factors

Factors such as fuel prices, weather conditions, and supply disruptions were not included in the analysis.

Scope Limitations

The project focuses on factory allocation and shipping optimization within the available dataset.

5.6 Future Scope

Future research can extend the project in several directions.

Real-Time Analytics

Integration with live operational systems.

Advanced Optimization Algorithms

Use of Genetic Algorithms and Reinforcement Learning.

Geographic Route Optimization

Incorporation of route mapping and geospatial analysis.

Cost-Based Simulation

Development of detailed transportation cost optimization models.

AI-Powered Decision Support

Implementation of intelligent recommendation systems using advanced AI techniques.

5.7 Conclusion

This project successfully developed a Factory Reallocation and Shipping Optimization Recommendation System for Nassau Candy Distributor using Business Analytics and Machine Learning techniques.

The study demonstrated how historical operational data can be transformed into actionable business intelligence through data preparation, exploratory analysis, predictive modeling, clustering techniques, recommendation systems, and dashboard visualization.

Among the evaluated machine learning models, Random Forest achieved the highest predictive accuracy and emerged as the preferred solution for lead-time estimation. K-Means Clustering provided valuable segmentation insights, while the recommendation engine identified high-impact optimization opportunities.

The Streamlit dashboard successfully translated analytical outputs into an interactive decision-support platform, enabling stakeholders to explore operational insights and recommendations.

Overall, the project demonstrates the potential of analytics-driven decision-making to improve shipping efficiency, support factory optimization initiatives, and enhance supply chain performance. The developed framework provides a scalable foundation for future optimization efforts and highlights the value of integrating machine learning into business operations.

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